THE SHEAF OF DIFFERENTIALS OF A MORPHISM OF ALGEBRAIC SPACES RESTRICTS TO THE RELATIVE DIFFERENTIALS ALONG AN ETALE COMMUTATIVE SQUARE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Etale morphisms of algebraic spaces induce equivalences of small etale topoi. Properties of Spaces, Lemma labelled spaces-properties-lemma-etale-morphism-topoi and Equation labelled spaces-properties-equation-restrict; invoked at lines 999-1003. Turns restriction of a sheaf on X_etale to U_etale into pullback along a_{small}, and likewise for b.
- Localisation of the module of differentials on ringed sites. Modules on Sites, Lemma labelled sites-modules-lemma-localize-differentials; invoked at lines 1003-1005. Computes Omega_{X/Y}|_{U_etale} as the module of differentials of the map of sheaves of rings induced by a_{small} and f_{small}.

Source reference. Original lemma 04BO: sources/stacks-chow/tags.txt, source lines 5659–5659. Navigation pointer for \ref{sites-modules-lemma-localize-differentials} used inside the counted span (line 1004); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LQ: sources/stacks-chow/tags.txt, source lines 4722–4722. Navigation pointer for \ref{spaces-properties-equation-restrict} used inside the counted span (line 1001); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03LP: sources/stacks-chow/tags.txt, source lines 4721–4721. Navigation pointer for \ref{spaces-properties-lemma-etale-morphism-topoi} used inside the counted span (line 1000); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Consider any commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow[\psi]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

where the vertical arrows are étale morphisms of algebraic spaces. Then

$$\Omega_{X/Y}|_{U_{étale}} = \Omega_{U/V}$$

In particular, if U, V are schemes, then this is equal to the usual sheaf of differentials of the morphism of schemes $U \to V$.

Proof. By Properties of Spaces, Lemma 03LP and Equation (03LQ) we may think of the restriction of a sheaf on $X_{étale}$ to $U_{étale}$ as the pullback by a_{small}. Similarly for b. By Modules on Sites, Lemma 04BO we have

$$\Omega_{X/Y}|_{U_{étale}} = \Omega_{\mathcal{O}_{U_{étale}}/a^{-1}_{small}f^{-1}_{small}\mathcal{O}_{Y_{étale}}}$$

Since $a^{-1}_{small}f^{-1}_{small}\mathcal{O}_{Y_{étale}} = \psi^{-1}_{small}b^{-1}_{small}\mathcal{O}_{Y_{étale}} = \psi^{-1}_{small}\mathcal{O}_{V_{étale}}$ we see that the lemma holds. □